DCT-Page: Frequency-Domain Page Proxies for Sparse KV Decoding

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Overview

DCT-Page is a training-free, decode-time sparse attention scheme that splits the KV cache into fixed-size pages, summarizes each page with a DCT-lowpass-IDCT proxy, and keeps only the top-k pages at full precision while leaving prefill — and thus the exact KV — unchanged.

Introduction

Key insight: attention keys are low-frequency signals

Our starting point is a structural observation about the attention key sequence itself: it lives almost entirely in the low-frequency band. Applying a DCT (Discrete Cosine Transform) to a 32-token page, the first 4 frequency coefficients already account for 85% of the total key energy. In other words, almost everything needed to route a query to the right page is concentrated in a handful of frequency components.

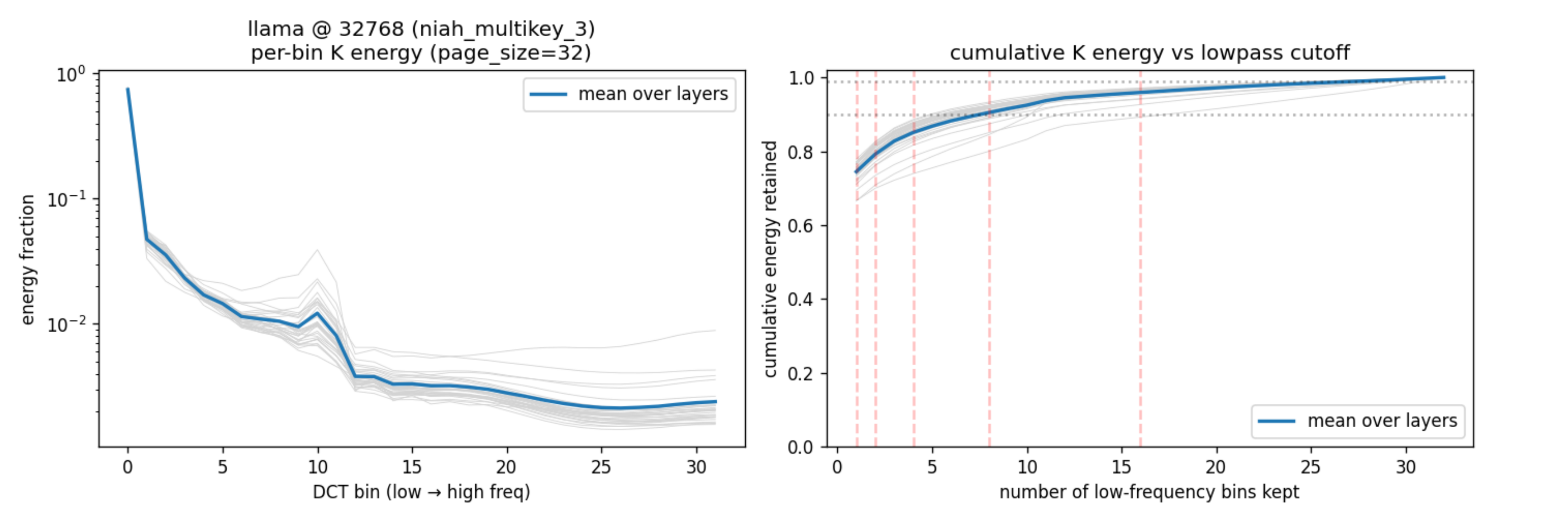

Fig. 1. Left: Per-bin DCT energy of K cache pages
Right: Cumulative K-cache energy vs lowpass cutoff

Acting on this, we summarize each page's keys with a DCT-lowpass followed by IDCT reconstruction — 32 tokens collapsed into 4 compressed tokens (8× compression) — and score pages by the query's inner product against this proxy. On a RULER dense-trajectory analysis, this proxy achieves 84.6% attention-mass recall, dramatically outperforming Quest's min/max scorer (62.9%) and approaching the oracle ceiling.

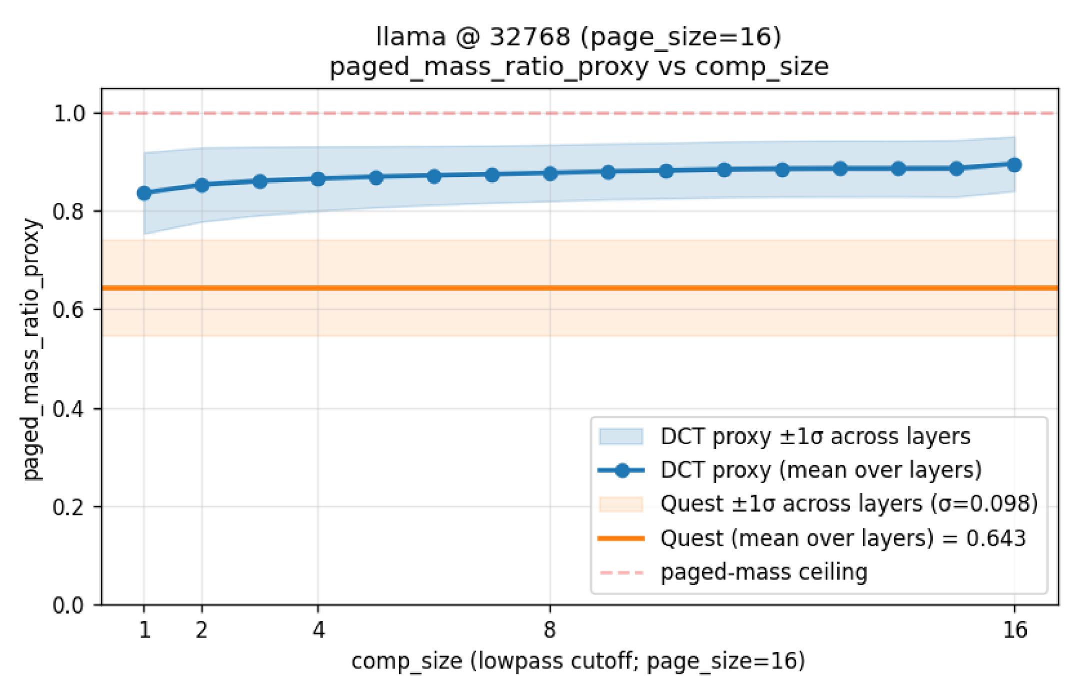

Fig. 2. Paged attention-mass recall vs. DCT lowpass cutoff

Contributions

Building on this insight, we present **DCT-Page**, a decode-time sparse page attention framework. Our contributions are:

- Training-free DCT-lowpass-IDCT page scoring — drop-in monkey-patch, no weight changes, no fine-tuning.
- Fused Triton kernels for scoring, top-k, KV assembly, and RoPE — turning theoretical sparsity into measured throughput.
- Comprehensive evaluation on RULER and LongBench v1/v2 with Llama-3.1-8B and Qwen3-8B, head-to-head against Quest, SeerAttention-R, and InfLLM, complemented by an attention-mass-recall analysis that pins down why DCT-Page works.

Methods

We apply a low-pass DCT filter to each page's key sequence, then score pages by the query's inner product against the reconstructed proxy. Top-k pages are read at full precision; the rest are dropped.

- **Proxy**: DCT keeps 4 low-frequency coefficients per 32-token page (8× compression)
- **Scoring**: max(q · compressed tokens), aggregated per GQA group with max

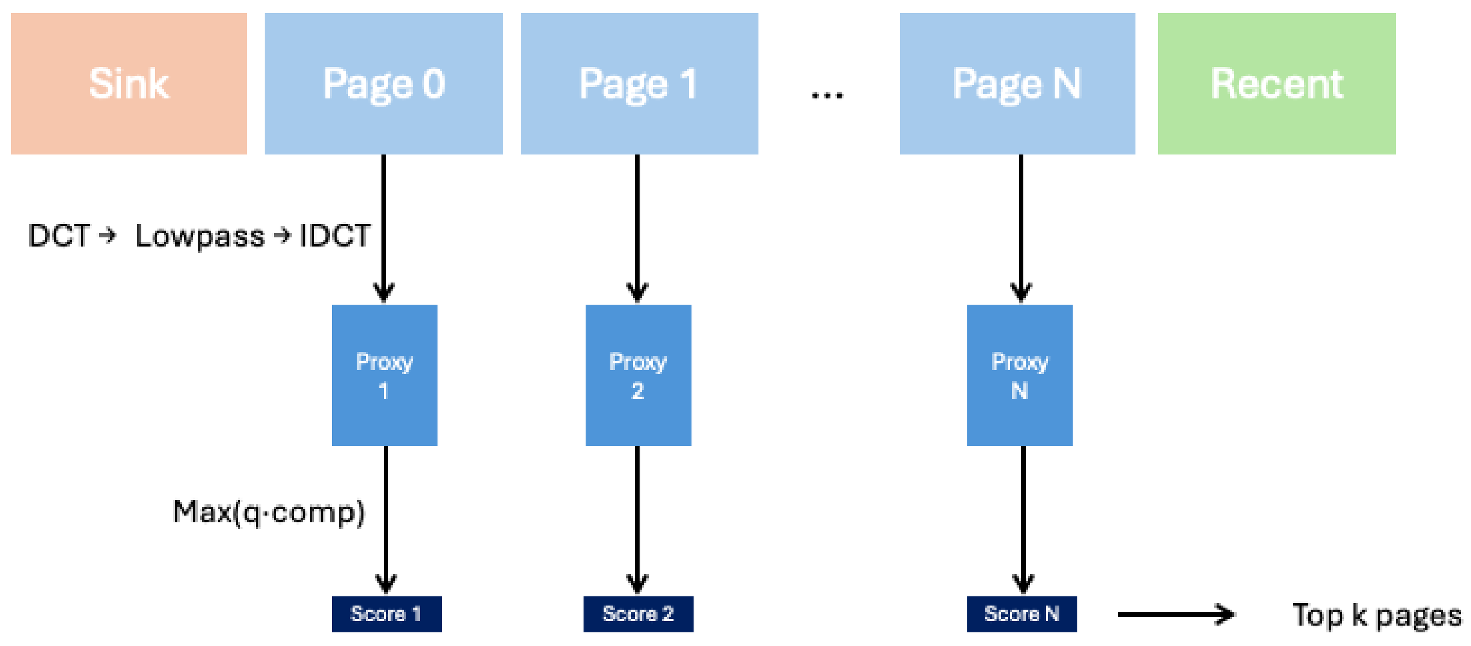

Fig. 3. Paged attention-mass recall vs. DCT lowpass cutoff

- **Selection**: sinlge sink page + top-k pages by score + 4 recent window pages
- **Implementation**: fused Triton kernels for scoring / top-k / KV-assembly / RoPE

Experiments

We evaluate DCT-Page on RULER (32K context, 13 synthetic long-context tasks), LongBench v1 (16 English tasks), and LongBench v2 (503 MC questions) with Llama-3.1-8B-Instruct and Qwen3-8B.

	NIAH-Single			NIAH-MultiKey			NIAH-Multi		Aggregation			QA		
Method	S1	S2	S3	MK1	MK2	MK3	MV	MQ	VT	CWE	FWE	QA1	QA2	Avg
Llama-3.1-8B-Instruct														
Full-KV	100.00	100.00	100.00	100.00	96.00	100.00	98.00	100.00	98.4	48.40	93.33	72.00	48.00	88.78
Quest	100.00	100.00	100.00	100.00	96.0	32.00	97.00	99.00	**97.60**	**30.00**	82.67	72.00	**48.00**	81.10
SeerAttn-R							*Qwen only*							
InfLLM	100.00	88.00	100.00	84.00	12.00	8.00	75.00	92.00	44.00	6.40	93.33	32.00	32.00	58.98
DCT-Page	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	96.00	28.80	**94.67**	**72.00**	44.00	**87.34**
Qwen3-8B														
Full-KV	100.00	100.00	100.00	96.00	88.00	96.00	95.00	100.00	96.00	86.40	93.33	56.00	48.00	88.83
Quest	100.00	100.00	100.00	96.00	88.00	8.00	90.00	90.00	98.40	**58.80**	84.00	48.00	48.00	77.32
SeerAttn-R	100.00	84.00	92.00	92.00	40.00	28.00	65.00	88.00	78.40	21.60	90.67	44.00	44.00	66.74
InfLLM							*Llama only*							
DCT-Page	**100.00**	**100.00**	**100.00**	**96.00**	**88.00**	**76.00**	**95.00**	**100.00**	**99.20**	34.80	**88.00**	**56.00**	**48.00**	**83.15**

Table 1. RULER accuracy (32K context). All sparse methods matched at 2048-token selected budget, block size 32, 4 reps/block. Best in **bold**.

- **DCT-Page leads the best sparse baseline (Quest) by +6.24 pts on Llama-3.1-8B-Instruct and +5.83 pts on Qwen3-8B**, while staying within 1.44 / 5.68 pts of Full-KV.

Benchmark	Full-KV	DCT-Page	Δ
LongBench v1	50.32	50.41	+0.09
LongBench v2	29.64	30.22	+0.58

Table 2. LongBench v1 (16 English tasks, average) and LongBench v2 (503 MC questions, overall accuracy) on Llama-3.1-8B-Instruct.

- **DCT-Page** stays within Δ ≤ 0.6 of Full-KV on both LongBench v1 and v2.

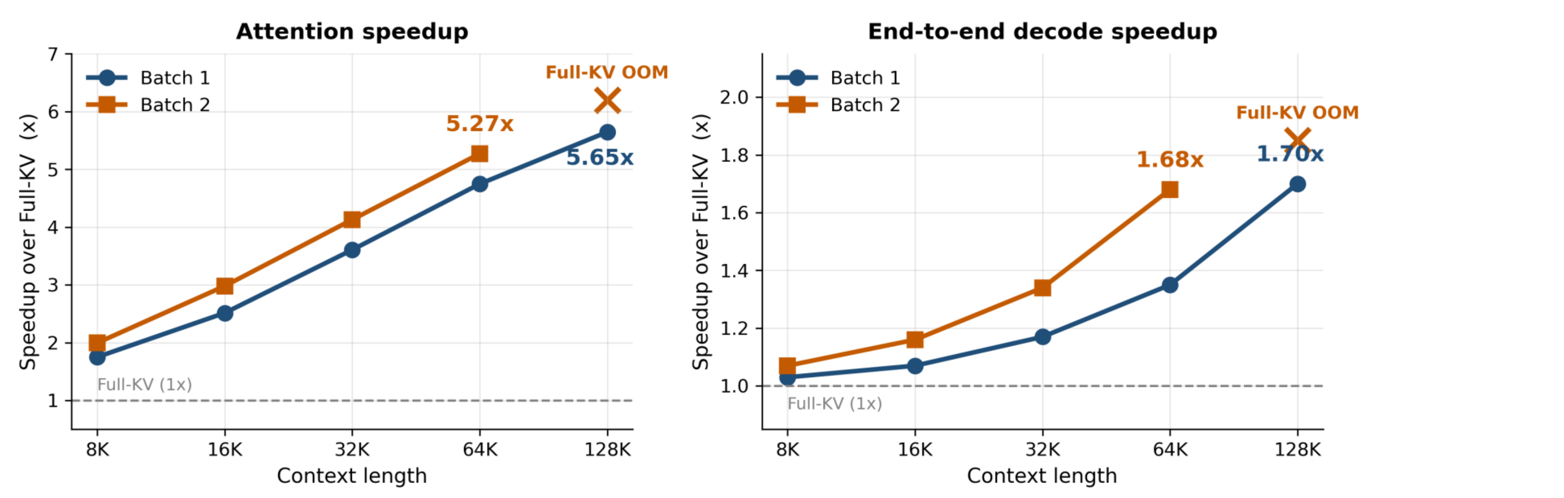

Fig. 4. Wall-clock decode speedup of DCT-Page over Full-KV on a single A6000 (Llama-3.1-8B-Instruct), measured with CUDA graphs. **Left**: attention-module speedup — up to **5.65× at 128K** (Batch 1) and **5.27× at 64K** (Batch 2). **Right**: end-to-end decode speedup including MLP and layer-norm — **1.70× at 128K** (Batch 1) and **1.68× at 64K** (Batch 2).

Conclusion

DCT-Page is a **training-free decode-time sparse attention** built on one observation — *attention keys are low-frequency*.

- **+6.24 / +5.83 pts** on RULER vs best sparse baseline (Llama / Qwen3)
- **84.6% attention-mass recall** (Quest: 62.9%) → routing quality drives accuracy
- **5.65× attention / 1.70× end-to-end** decode speedup at 128K

Future: closing the Qwen3 Full-KV gap, extending to prefill, scaling to larger models.

Reference

[1] Xiao et al. InfLLM: Training-Free Long-Context Extrapolation for LLMs with an Efficient Context Memory. NeurIPS 2024.
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[3] Gao et al. SeerAttention-R: Sparse Attention Adaptation for Long Reasoning. arXiv:2506.08889, 2025.
[4] Kai et al. FreqKV: Key-Value Compression in Frequency Domain for Context Window Extension. ICLR 2026.